ANALYTICAL CUMULATIVE EXAM
April 22, 1995

(20 pts) 1. Explain in general how pH measurements are made using a glass pH electrode. Explain how the potential across the membrane that lies between a reference and analyte solution can yield information about the pH of the analyte solution. Describe the structure and features of the glass membrane in detail. Include a discussion of the hygroscopicity of the glass membrane in terms of the formation of two "phases" in the membrane. What species conduct in the two different "phases"? Write out the chemical equilibria that occur for Na+ and H+ at the glass/solution interface.

(20 pts) 2. Derive the following expression for the determination of concentration (actually it is the activity, *a*) of X^{n+} from the measured potential, E_{cell}, across the membrane. The analyte solution of X^{n+} is on one side of the membrane and a reference solution of known concentration or activity of X^{n+} is on the other side of the membrane.

$$pX = -\log a_X = -(E_{cell} - K)/(0.0592/n) \qquad (1)$$

where *n* is the number of moles of electrons transferred per mole of analyte, K is constant that includes and constant terms such as reference solution concentrations or activities and junction potentials, and a_X is the activity of the species X^{n+}.

To start the derivation, assume that E_{cell} is only dependent on the *difference* in potentials that are established on each side of the membrane. Use the Nernst Equation to express the potential as a function of concentration for each side of the membrane.

$$E = E^\circ + (0.0592/n)\log(a_O/a_R)$$

where E° is the standard reduction potential. Note that a_O is the activity of the oxidized form of the species and a_R is the activity of the reduced form of the species. Assume that the oxidized and reduced forms of the analyte that can be used in the Nernst Equation are related by the following reaction

$$X^{n+} + ne^- \text{--->} X(s).$$

(20 pts) 3. Use the above Equation (1) in the following problem. The cell

SCE||H+ (a = x)|glass electrode

has a potential of **0.2094** V when the solution between the glass electrode and the SCE reference electrode is a buffer of **pH 4.006**. The following potentials are obtained when the buffer is replaced with unknowns:

(a) -0.3011 V
(b) +0.1163 V

Calculate the pH and the hydrogen ion activity of each unknown.

(20 pts) 4. Assuming an uncertainty of ±0.002 V in the junction potential, what is the range of hydrogen ion activities within which the true value might be expected to lie?

(20 pts) 5. A recent article in *Analytical Chemistry* (**1995**, *67*, 1346-1356) by Chang Wei, Allen J. Bard, Geza Nagy, and Klara Toth describes the use of a scanning liquid membrane microelectrode to image the concentrations of ions spatially.

(a) The theory behind measurements of analyte activies using liquid membrane ion-selective electrodes is similar to that of glass electrodes. However, the structure of the membrane is different in that it contains a hydrophobic liquid with ion-exchanging molecules. In this sense, it is more versatile than a glass electrode in that any of a large number of known ion-exchanging molecules can be incorporated into the liquid membrane.
Suggest *two* kinds of ion-exchanging molecules that could be used in a liquid membrane ion-selective electrode. (Think back to organic chemistry and ways in which to solubilize metal ions in organic phases.) Describe the kinds of selectivity for different ions that you would expect for your ion-exchangers.

(b) Micro-liquid membrane ion-selective electrodes are available commercially. Discuss some applications for miniaturized liquid membrane ion selective electrodes. Suggest difficulties or limitations in making ion activity measurements on microscopic scales and for the applications that you describe.

(c) Some liquid membranes are highly sensitive to potassium ions even in the presence of very large amounts of sodium ions. Suggest possible applications in which this kind of selectivity would be useful.